

Mobility-Budget-Constrained Submodular Maximization for Distributed Semantic Multi-Robot Exploration

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Abstract—Multi-robot exploration under communication and mobility constraints requires balancing information gain with reachability. We present a distributed semantic exploration framework that combines Dirichlet mapping, budget-constrained trajectory selection, and RAG coordination. Each robot clusters frontiers, computes terrain- and heading-aware least-cost paths using a single budget-limited Dijkstra search, and selects feasible paths by marginal information gain while reducing redundant sensing. We establish guarantees under full and limited communication. Gridworld ablations show improved coverage and semantic accuracy with lower entropy and traversal cost than closest-frontier exploration. Full-system simulations further demonstrate conflict-free coordination and concurrent commitments across disjoint neighborhoods without reducing coverage.

I. INTRODUCTION

Multi-robot exploration enables teams to map unknown environments for applications such as search and rescue, reconnaissance, and planetary exploration [1]. Robots must coordinate their actions while balancing information acquisition against resource limitations. Semantic maps require greater communication bandwidth than binary occupancy maps, potentially delaying and degrading coordination. Resource-aware semantic information acquisition is therefore essential (Fig. 1).

We propose a distributed semantic exploration framework in which each robot maintains a Dirichlet semantic map fused from source-attributed records exchanged with neighbors. Frontier regions are clustered into candidate targets, and a single budget-limited Dijkstra search over a heading-augmented graph returns least-cost paths under semantic traversal and heading-change costs. Robots de-conflict trajectories through the Resource-Aware distributed Greedy (RAG) protocol [2] using local marginal-gain comparisons.

The contributions of this work are:

- 1) A budget-constrained submodular formulation that maximizes semantic information gain subject to an exact, edge-additive mobility budget incorporating terrain-dependent traversal and heading-change costs.
- 2) RAG-based distributed coordination with approximation guarantees under limited communication.
- 3) A complete multi-robot implementation in MATLAB and high-fidelity ground-robot simulations.

II. RELATED WORK

Communication-constrained multi-robot exploration includes distributed SLAM [3], low-bandwidth planning [4], and dynamic relay architectures [5]; deployments in extreme environments further motivate resource-efficient distributed systems [6]–[8]. Active exploration methods select

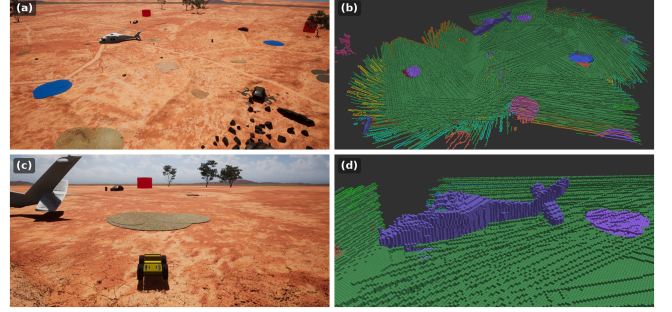


Fig. 1. Distributed semantic multi-robot exploration in HERCULES. (a) The outdoor arena, containing terrain patches of differing traversability alongside structures and scattered debris. (b) The 3-D semantic voxel map accumulated online by the robot team, with voxels colored by inferred semantic class. (c) A Husky ground robot exploring the arena. (d) Close-up of a reconstructed aircraft and an adjacent terrain patch, recovered from the robots’ synchronized depth and segmentation observations.

trajectories using information-theoretic objectives [9]–[12]. Bayesian multi-class mapping maintains per-cell categorical distributions and provides a closed-form semantic mutual-information bound [13], supporting single-robot semantic exploration [14] and multi-robot distributed optimization [15]. However, these methods do not jointly address terrain-dependent mobility budgets and limited inter-robot communication. RAG [2] enables communication-efficient distributed submodular maximization with topology-dependent guarantees. We combine a Dirichlet-categorical semantic grid with terrain-aware mobility constraints and RAG-based coordination under limited communication.

III. PROPOSED METHOD

A. Semantic Map Representation and Fusion

Following [13], the map M is a 2-D grid where each cell j carries a semantic probability vector $\theta_j = (\theta_j^1, \dots, \theta_j^K)$ over K classes. Given one-hot semantic measurements and a Dirichlet prior $\text{Dir}(K, \alpha_0)$, the posterior parameters of cell j are

$$\alpha_j^k = \alpha_0^k + \sum_{x_i} k(x_i, x_j) y_i^k, \quad (1)$$

where the Gaussian kernel weight $k(x_i, x_j)$ encodes that closer observations are more reliable. The posterior mean $E[\theta_j^k] = \alpha_j^k / \sum_k \alpha_j^k$ and the differential entropy $H(\text{Dir}(\alpha_j))$ [16] are first-class quantities of the map. The mutual information between the map and observations Z

over a perceptual field $\mathcal{O}$ is

$$I(M; Z) = \sum_{j \in \mathcal{O}} \left[H(\text{Dir}(\alpha_j^{\text{prior}})) - H(\text{Dir}(\alpha_j^{\text{post}})) \right]. \quad (2)$$

During planning, future observations are unknown; each robot therefore evaluates the conditional expected reduction in entropy given its observations so far.

Each robot stores occupancy and semantic records by observation source. Let $\alpha^{(i,s)}$ be source s 's semantic record stored by robot i , and let $\mathcal{S}_i$ be the available sources. Its accumulated observation count is

$$C^{(i,s)} = \sum_j \sum_{k=1}^K \left(\alpha_j^{k,(i,s)} - \alpha_0^k \right). \quad (3)$$

When neighbors i and r communicate, they exchange their source-attributed records. For each source s , robot i retains the record with the larger count:

$$\alpha^{(i,s)} \leftarrow \begin{cases} \alpha^{(r,s)}, & C^{(r,s)} > C^{(i,s)}, \\ \alpha^{(i,s)}, & \text{otherwise.} \end{cases} \quad (4)$$

Records remain indexed by their original source, and robot i forms its fused map as

$$\alpha_j^{k,\text{fused},(i)} = \alpha_0^k + \sum_{s \in \mathcal{S}_i} \left(\alpha_j^{k,(i,s)} - \alpha_0^k \right). \quad (5)$$

Replacing, rather than summing, alternative versions of the same source record prevents double counting while allowing information to propagate through successive encounters. Under limited communication, each robot plans using the most complete records it has received.

B. Budget-Constrained Problem Formulation

A trajectory $a_i = (p_1^i, \dots, p_{L_i}^i)$ is an ordered cell sequence, and its observation set $Z_i(a_i)$ is the union of sensor-footprint cells swept along it. Each cell p has a semantic traversal cost $c_{s,p} = \sum_k E[\theta_p^k] S_k$, with $S \geq 0$ the per-class cost vector (e.g., $S_{\text{road}} < S_{\text{grass}} < S_{\text{mud}}$). The augmented trajectory cost adds a heading-change penalty over interior waypoints,

$$c_s(a_i) = \sum_{p \in a_i} c_{s,p} + \lambda \sum_{\ell=2}^{L_i-1} (\delta_\ell^i / \pi)^2, \quad \lambda \ll \min_k S_k, \quad (6)$$

where δ_ℓ^i is the turning angle at waypoint ℓ . The multi-robot exploration problem is then

$$\max_{\mathbf{a} \in \mathcal{A}} I\left(M; \bigcup_{i=1}^n Z_i(a_i)\right) \quad \text{s.t.} \quad c_s(a_i) \leq B \quad \forall i, \quad (7)$$

a monotone submodular maximization under per-robot knapsack constraints. Since c_s appears only in the constraint, the monotonicity and submodularity of the objective are unaffected by the augmented cost. Because costs are additive over the fixed least-cost path to each candidate rather than optimized over orderings, the problem retains knapsack rather than orienteering [17] structure.

Algorithm 1 Budget-Constrained Semantic Exploration with RAG Coordination (robot i , round t)

Require: fused map $M_i^{(t)}$, pose $x_i^{(t)}$, budget B , class costs S , neighborhood N_i

Ensure: committed trajectory a_i^*

- 1: Extract and cluster frontier cells of $M_i^{(t)}$ into candidate targets
 - 2: Run one budget-limited Dijkstra search from $x_i^{(t)}$ over the heading-augmented graph with edge costs realizing (6)
 - 3: $\tilde{\mathcal{A}}_i^{(t)} \leftarrow$ least-cost paths with $c_s(a) \leq B$; $C \leftarrow \emptyset$
 - 4: **repeat**
 - 5: Compute $\Delta(a \mid \psi_i^{(t)}, C)$ for all $a \in \tilde{\mathcal{A}}_i^{(t)}$
 - 6: $a_i^* \leftarrow$ better of $\arg \max_a \Delta / c_s(a)$ and $\arg \max_a \Delta$;
 $g_i \leftarrow \Delta(a_i^*)$
 - 7: Exchange g_i with undecided robots in N_i
 - 8: **if** g_i largest among undecided robots in N_i (ties by index) **then**
 - 9: Commit a_i^* and broadcast to N_i
 - 10: **else**
 - 11: Receive newly committed trajectories into C
 - 12: **end if**
 - 13: **until** robot i has committed
 - 14: Execute a_i^* ; update $\alpha^{(i)}$ via (1); exchange source-attributed records, update via (4), and fuse via (5)
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C. Distributed Planning with RAG Coordination

At round t , robot i clusters frontier cells—unexplored cells adjacent to known traversable space—into candidate targets. One budget-limited Dijkstra search over a heading-augmented graph returns the least-cost path from the robot to every candidate. Graph nodes are cell-heading pairs, and edge costs realize (6). Search pruning yields the exact feasible candidate set

$$\tilde{\mathcal{A}}_i^{(t)} = \{a : c_s(a) \leq B\}.$$

Selection uses the modified cost-benefit greedy rule: among feasible candidates, compute the conditional expected marginal gain $\Delta(a \mid \psi_i^{(t)}, C)$ given observations so far and the trajectories C committed by neighbors, and return the better of the ratio maximizer $\arg \max_a \Delta / c_s(a)$ and the singleton maximizer $\arg \max_a \Delta$. Coordination follows the RAG protocol [2]: undecided robots exchange their best marginal gains with neighbors; a robot whose gain is largest among its undecided neighbors commits and broadcasts its trajectory, and remaining robots recompute marginals on the residual objective. Under partial communication, robots in disjoint neighborhoods commit concurrently in the same communication round. Algorithm 1 summarizes one planning round.

D. Approximation Guarantees

Let OPT denote the centralized solution to (7) using all fused-map information available at round t . Conditional independence of observations given the map, as in the

Dirichlet-categorical model, makes the objective monotone and submodular [18], [19]. Full derivations are deferred to the journal version.

Theorem 1 (Per-round, single robot): For monotone submodular f under a knapsack constraint, modified cost-benefit greedy returns S satisfying $f(S) \geq 0.427f(\text{OPT})$ [20]. Sviridenko’s partial enumeration achieves $(1 - 1/e)$ at higher computational cost [21].

Theorem 2 (Per-round, multi-robot): With sequential commitment, full information exchange, and an α -approximate marginal solver, the joint solution satisfies $f(\mathbf{a}) \geq \frac{\alpha}{1+\alpha} f(\text{OPT})$.

Thus, the implemented modified greedy ($\alpha = 0.427$) guarantees approximately 0.299 of the centralized per-round optimum. Under limited communication, RAG [2] gives

$$f(\mathbf{a}^{\text{RAG}}) \geq \frac{1}{1 + \kappa_f} \left[f(\text{OPT}) - \kappa_f \sum_i \text{coin}_{f,i}(N_i) \right], \quad (8)$$

where κ_f is the curvature of f , and $\text{coin}_{f,i}$ measures sensing overlap between robot i and robots outside N_i . The degradation vanishes when non-neighbor footprints do not overlap, recovering the fully connected case when $\text{coin}_{f,i} = 0$.

Map evidence propagates transitively through source-attributed records, whereas trajectory commitments are conditioned only on current neighbors. Thus, limited communication delays map evidence but confines coordination to N_i ; the degradation term captures overlap with non-neighbors’ actions.

At the mission level, conditional independence also makes the objective adaptive monotone and adaptive submodular. Consequently, greedy replanning achieves at least $(1 - 1/e)$ of the optimal adaptive policy’s expected total information gain [22]. The per-round bounds compare one allocation with the best fixed allocation for that round, whereas the mission-level bound compares the complete policy with a clairvoyant adaptive planner.

IV. EXPERIMENTS

A. Semantic Gridworld Ablation

We evaluate the framework in a semantic gridworld containing six traversable terrain classes—dry soil, grass, dirt, sand, mud, and gravel—and obstacles. Five robots explore the initially unknown environment with sensor range $r_s = 10$ cells, communication range $r_c = 20$ cells, normalized mobility budget $B = 1$, and a 5-s receding execution horizon. The simulated semantic sensor reports the correct semantic class with probability 0.6 and distributes the remaining probability uniformly among the incorrect classes.

We compare four modes:

- 1) *Shortest Distance Frontier*: geometric nearest-frontier selection without information gain or coordination.
- 2) *Geometric Cost + Semantic IG*: geometric path costs with semantic IG and RAG coordination.
- 3) *Semantic Cost + Occupancy IG*: semantic path costs with occupancy IG and RAG coordination.

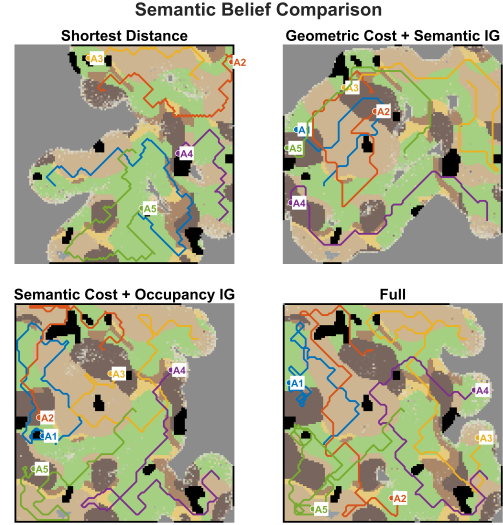


Fig. 2. Final fused semantic maps for a representative paired run after 300 s. Colored lines show robot trajectories, black cells are obstacles, and gray cells are unexplored.

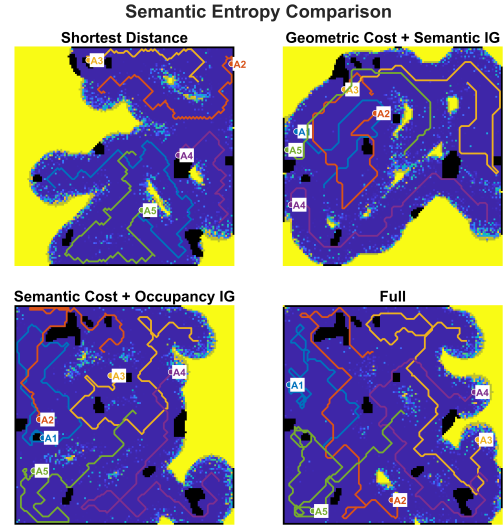


Fig. 3. Final semantic-entropy maps for the same representative run. Darker cells indicate lower uncertainty, while yellow regions are unobserved.

- 4) *Full*: semantic path costs and semantic information gain with RAG coordination.

We conduct 10 paired 300-s runs, using the same seed and initial robot positions across modes within each run. Coverage is measured over traversable cells, whereas mIoU and mean semantic entropy are evaluated over the entire map. Route efficiency is the standardized semantic traversal and heading-change cost divided by the team’s total geometric distance.

Table I reports the mean results, while Figs. 2 and 3 show a representative run. Relative to Shortest Distance, both partial ablations improved coverage, mIoU, and entropy. The Full method achieved the best mean mapping performance on all three map-quality metrics, and the terrain-aware modes achieved lower route costs than the geometric-cost modes.

TABLE I
SEMANTIC GRIDWORLD ABLATION AFTER 300 S OVER 10 PAIRED RUNS (MEAN $\pm$ STANDARD DEVIATION).

Mode	Coverage (%)	mIoU (%)	Entropy (nats)	Route Efficiency (cost/distance)
Shortest Distance Frontier	79.00 $\pm$ 9.52	64.13 $\pm$ 10.00	0.4481 $\pm$ 0.1722	0.1263 $\pm$ 0.0099
Geometric Cost + Semantic IG	87.58 $\pm$ 5.55	73.06 $\pm$ 6.57	0.2835 $\pm$ 0.1044	0.1282 $\pm$ 0.0045
Semantic Cost + Occupancy IG	88.72 $\pm$ 6.26	73.62 $\pm$ 8.74	0.2548 $\pm$ 0.1127	0.1036 $\pm$ 0.0065
Full	91.32 $\pm$ 6.19	77.48 $\pm$ 7.56	0.2003 $\pm$ 0.1087	0.1009 $\pm$ 0.0039

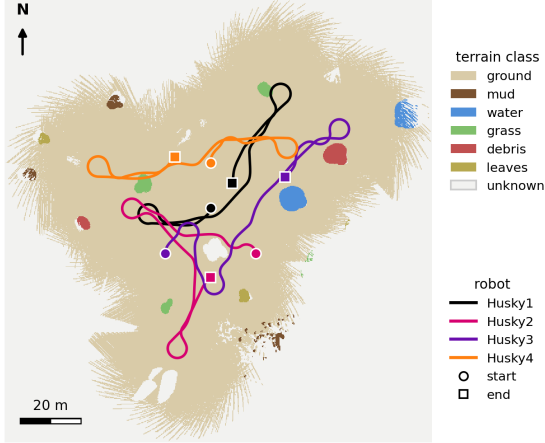


Fig. 4. Fused semantic ground map and robot trajectories after a 600 s coordinated exploration run with four ground robots in HERCULES. Circles mark start poses and squares mark final poses.

B. Full-System Evaluation

We evaluate the complete framework in HERCULES [23], an open-source Unreal Engine 5 simulation framework for heterogeneous multi-robot teams in photorealistic outdoor environments, with a team of Husky ground robots. Each robot builds a 3-D semantic voxel map from synchronized depth, segmentation, and pose, derives a 2-D Dirichlet semantic grid by per-column clearance analysis, and runs the frontier, planning, and coordination stack described above. Sensor data are collected in lockstep so that depth, segmentation, and pose share a single simulation timestamp, and a communication broker enforces the configured topology on all exchanged source-attributed map records, bids, and commitments. Fig. 4 shows a representative run.

Coordination. Across three 600 s runs per mode with four robots under full communication, all coordinated modes eliminate target conflicts (mean same-target robots per round approximately zero, against 2.02 for independent selection). Coverage and certainty gained per unit standardized cost do not separate the modes at this sample size: all 95% confidence intervals overlap, and independent selection attains the highest certainty per unit cost precisely because redundant re-observation is cheap, while covering the least new area. Reporting new-area efficiency alongside certainty is therefore necessary to interpret either. Against the two literature coordination rules reimplemented in the same stack, semantic-RAG attains higher certainty per unit cost than both

TABLE II
FULL-SYSTEM COORDINATION RESULTS IN HERCULES: FOUR ROBOTS, 600 S RUNS, FULL COMMUNICATION.

Method	Cert. /cost	Area /cost	Cov. (k)	Confl. /rnd
Independent	2.06	8.80	141.8	2.02
Auction (SSI)	1.88	9.32	150.8	≈ 0
Voronoi [†]	1.78	11.13	160.2	≈ 0
NBV-3DMR ^{†*}	1.47	11.80	165.2	≈ 0
ROAM-overlap [*]	1.54	9.03	146.3	≈ 0
Semantic-RAG [†]	1.73	9.34	145.0	≈ 0

Cert./cost: certainty gained per unit standardized cost. Area/cost: new area per unit cost. Cov.: fused known cells (thousands). Confl.: mean same-target robots per round. $n = 3$ per mode; $*n = 1$. All 95% confidence intervals overlap on the efficiency columns, so no values are bolded. [†]One or more runs lost a robot to a simulator terrain-collision defect; such runs travel less and bias the per-cost columns upward.

NBV-3DMR and the ROAM-style overlap penalty, though both were single runs and the NBV-3DMR run lost a robot. Table II reports the coordination results.

Communication topology. Under a chain topology, in which the two end robots cannot communicate, semantic-RAG retains the coverage of the fully connected condition while committing concurrently in disjoint neighborhoods, averaging 2.77 commitments per coordination round, and its selections capture 84% of the gain available to a fully connected shadow planner. Coordination therefore trades a bounded selection sacrifice for concurrency at no measurable coverage cost.

V. CONCLUSION

We presented a distributed semantic multi-robot exploration framework combining Dirichlet mapping, budget-constrained submodular trajectory selection, and RAG coordination, with approximation guarantees under limited communication. Gridworld ablations show that jointly using semantic information gain and traversal costs improves mapping performance and route efficiency, while full-system simulations show that coordinated selection eliminates target conflicts at no measurable coverage cost, and that efficiency metrics do not separate the coordination rules at this sample size. The mobility budget did not bind in this arena, so its benefit is not exercised here; future work tightens the budget relative to environment scale, expands the semantic classes, increases the number of trials, and validates the framework on ground robots.

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